Erdheim-Chester disease

A rare histiocytosis. Around 800 cases reported in the literature.

Foamy histiocytes accumulate in bone, in the tissue around the kidneys, along the aorta and in the brain. What the patient feels depends entirely on where they settle, which is part of why it is difficult to recognise.

Nearly all cases carry an acquired mutation in one pathway



BRAF V600E is the commonest. It is acquired in the histiocytes themselves, not inherited.

Which is why it became treatable

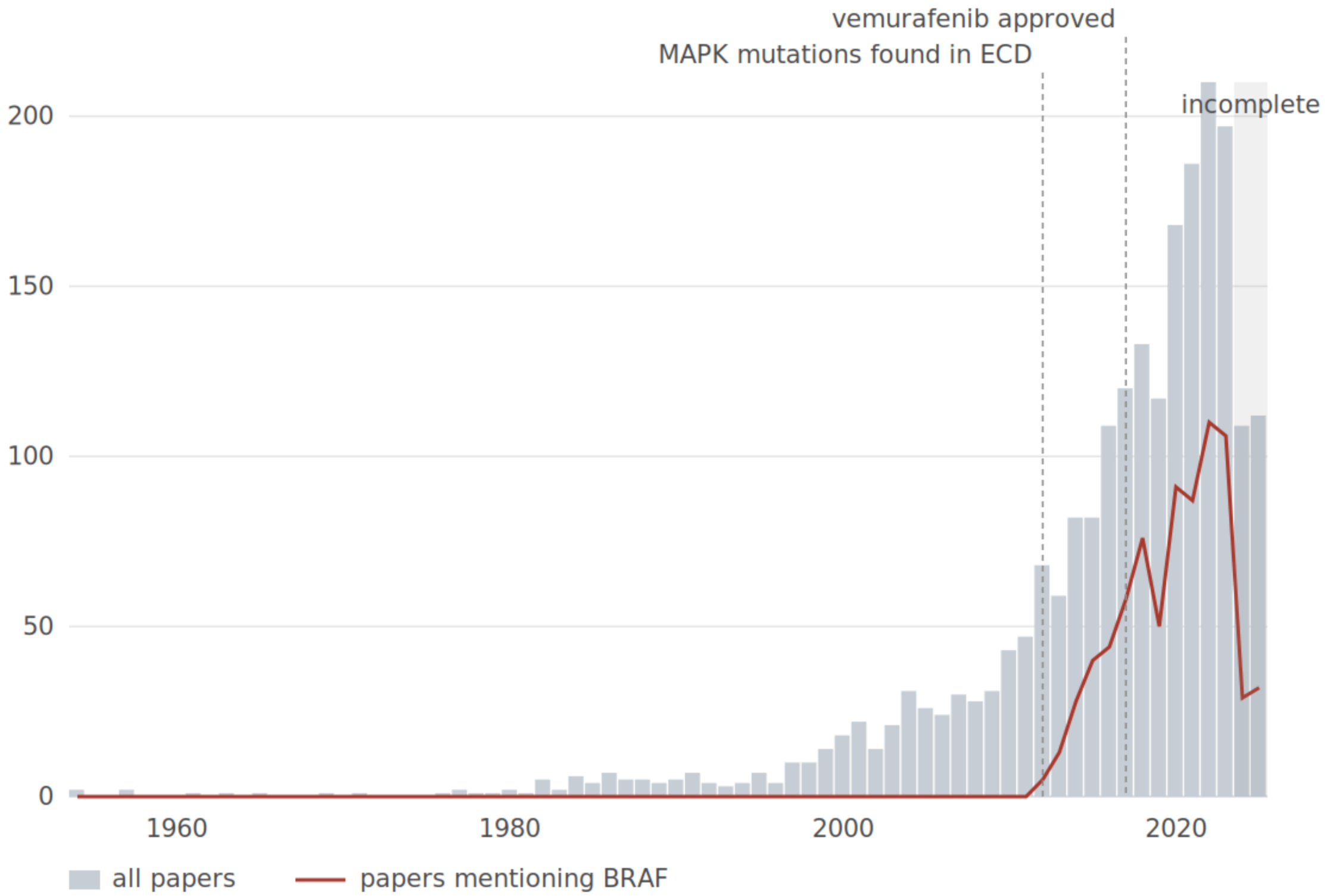
Vemurafenib, a BRAF inhibitor, was approved for it in November 2017.

Almost everything known about this disease sits in roughly two thousand papers.

Most of them are recent, and that is not a coincidence.

Fifty years of case reports, then a gene

2,288 papers, 1954 to 2026, across 874 journals.



4 papers in the whole of the 1950s. 870 between 2020 and 2024. The red line is papers mentioning BRAF: flat at zero for five decades, then tracking the entire field.

The shaded years are not a decline — they are not fully annotated yet.

So: about two thousand papers, most of them written since the gene was found. How would you go and find them all?

How you find those papers today

You search PubMed for the name. That search reads a paper's title, its abstract, its keywords and its index terms. It does not read the body of the paper.

And the name is not written one way

`Erdheim-Chester disease`	`Erdheim Chester disease`
`ECD`	`Erdheim-Chester’s disease`
`Chester-Erdheim disease`	`Erdheim-Chester syndrome`
`lipid granulomatosis`	`Erdhiem-Chester disease`

What a search for the name returns

1,600 papers

Four of these would not be matched by a search for “Erdheim-Chester”: ECD on its own, lipid granulomatosis, the reversed Chester-Erdheim, and the misspelling.

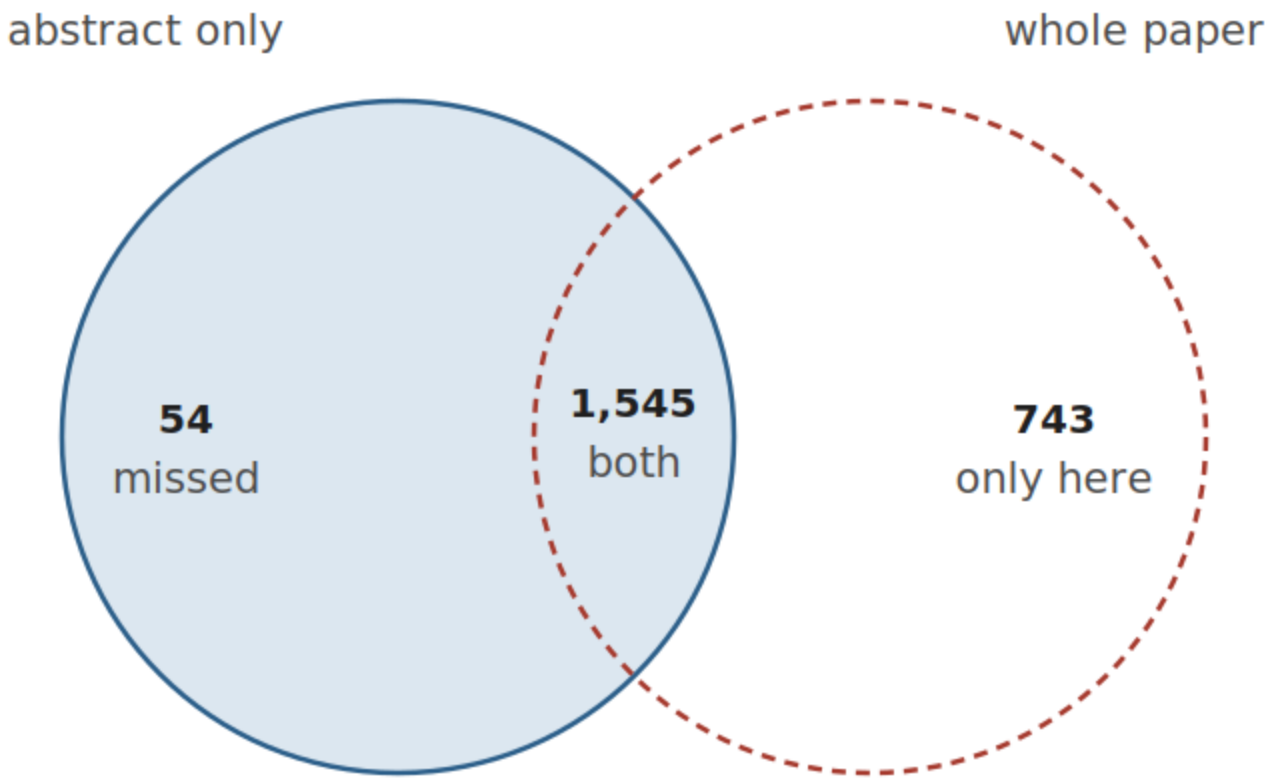
In practice most papers spell the name out at least once, so the variants alone cost less than they look.

The larger gap is structural

A name search never reads past the abstract.

Reading the whole paper

Not just the title and the abstract.



Where the 743 came from

113 of 120	sampled papers never name the disease in title or abstract	

And they sit where the body text can be read

701 of 743	of the surplus papers are in PubMed Central	**94%**
236 of 600	of the papers both systems find are	**39%**

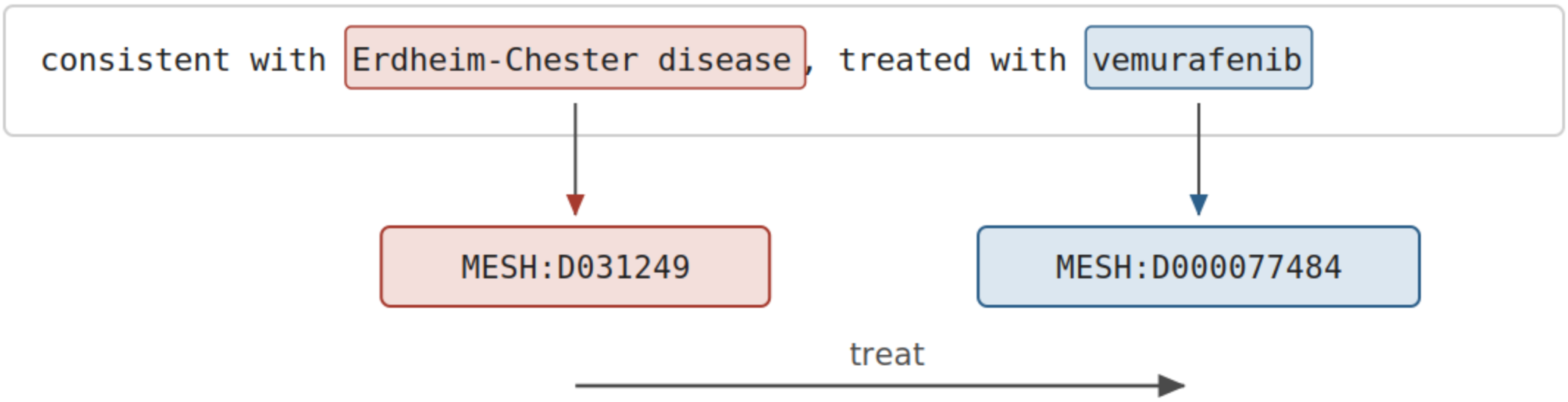
And this part is not clever

A plain text search of full-text articles in PubMed Central returns 2,349. The extra papers come from reading more of each paper. Nothing more.

So the count is not the interesting part, and it is not what the annotation was for. The next two pages are.

How a machine decides what a name means

Three steps, and only the second one is hard.



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1 **Find the mention**

Spot that a run of words is naming a disease, or a drug.

2 **Resolve it**

Decide which concept those words refer to, and attach a standing identifier. This is the step that makes counting possible, and the only step that can be wrong about what a name means.

3 **Link them**

Record that the sentence asserts a relation between the two.

Step 1 is one system. Step 2 is a different system for each kind of entity, and step 3 is a third. Diseases and genes never share a resolver.

What that buys: structure, not papers

A text search hands you a pile of papers to read. Resolving every mention to an identifier hands you a table you can count.

Genes named

BRAF	806
CD68	593
CD1a	578
S100	423

Drugs reported as treating it

vemurafenib	83
cladribine	28
dabrafenib	18
cobimetinib	17

Conditions these papers also discuss

Langerhans cell histiocytosis	751
non-Langerhans cell histiocytosis	690
Rosai-Dorfman disease	377

Those three are the differential diagnosis. They were not part of any query - they came out of the counting.

A search returns the papers that mention the disease. Once every mention carries an identifier, you can ask which gene, which drug, which year - and get a number back.

Which puts all the weight on one step: deciding what each name refers to.

Two diseases, one surname

Jakob Erdheim described two unrelated conditions.

Erdheim-Chester disease

a histiocytosis

bone, kidneys, CNS

`MESH:D031249`

Erdheim-Gsell necrosis

a degeneration of artery wall

aorta, iliac, basilar

`MESH:C536230`

Of 40 papers whose only evidence was an ambiguous form of the name:

35 were the aortic disease,
filed as the histiocytosis

Titles now labelled as a rare histiocytosis

`Cystic medionecrosis of the aorta (Gsell-Erdheim syndrome)`

`Aortic rupture in idiopathic Gsell-Erdheim medionecrosis`

`Aortic lesion in Marfan syndrome: cystic medial degeneration`

That last one is a Marfan syndrome paper.

The right answer was already there

MeSH has a concept for the aortic disease. PubTator3 knows it, and uses it 618 times elsewhere in this same corpus.

Here it was not used. Each mention carries one identifier and no measure of doubt.

For scale, on this disease

949 of 965	papers expert indexers had marked were found	**98.3%**
35 of 2,288	papers labelled here were a different disease	**1.5%**

What it does when it does not know

The format can say “found it, could not place it”. Some entity types use it.

	Annotation rows	Left unplaced	Two or more candidates
Diseases	168,305,967	0	0
Chemicals	146,957,635	10,749,370	0
Genes	74,519,899	0	310,917
Mutations	8,135,311	0	0

Two of the four types can express doubt

Chemicals leave 7 in every 100 rows unplaced. Genes return two candidates 310,917 times. Diseases place all 168 million.

What a chemical looks like when it gives up

Ca2+	58,260	cAMP	25,434	O2	20,763
ROS	14,978	ketamine	12,039	PM2.5	8,593

Ions, radicals and mixtures, mostly. Some, like ketamine, are real misses.

Diseases are where the names collide. Diseases are the one type that always answers.

Sources and method

Wei C-H, Allot A, Lai P-T, et al. PubTator 3.0: an AI-powered literature resource. Nucleic Acids Research 52(W1):W540-W546, 2024. Data: NCBI PubTator3 bulk files, build 2026-08-17, public domain. Publication dates: NCBI E-utilities.

Goyal G, Heaney ML, Collin M, et al. Erdheim-Chester disease: consensus recommendations for evaluation, diagnosis and treatment in the molecular era. Blood 135(22):1929-1945, 2020.

Diamond EL, Subbiah V, Lockhart AC, et al. Vemurafenib for BRAF V600-mutant Erdheim-Chester disease and Langerhans cell histiocytosis. JAMA Oncology 4(3):384-388, 2018.

How the numbers were produced

438,013,195 annotation rows parsed from five files; 0 unreadable.
Papers selected by MeSH descriptor D031249. Comparison counts from PubMed E-utilities: 1,600 for the plain name search, 965 for the MeSH descriptor alone, and 2,349 for a full-text search of PubMed Central.
Full-text reach tested by PubMed Central membership against a 600-paper control.
The eponym errors were classified from titles and abstracts, then re-checked against the live PubTator3 API, where 5 of 6 reproduced.
Unresolved counts are rows whose identifier is '-'; multi-candidate rows are identifiers that are a list of ids. Mutation identifiers are a compound key and are excluded from that test. MESH:C536230 usage counted across all 168m rows.

Step 1 is AIONER; step 2 is TaggerOne for diseases, GNorm2 for genes, NLM-Chem for chemicals and tmVar3 for mutations; step 3 is BioREx.

Code and full results

`github.com/GovardhanKS/pubtator3-rare-disease-pipeline`

Counts are of papers, never of patients. Published patient frequencies for BRAF V600E in this disease are around 50% to 60% (Goyal et al.).

Schematic prepared with AI assistance. Content based on the sources above.